CONNEXUS black sea tech forum | 2026

From Data to Decisions

Sofia's Digital Twin and City Living Lab in Practice

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GATE
big data for smart society

ICT CLUSTER VARNA

BSMEPA
BULGARIAN SMALL AND MEDIUM ENTERPRISES PROMOTION AGENCY

BAPHA

BRAIN++
AI FACTORY

ARC FUND



A QUICK QUESTION

How many of you are familiar with what a living lab is?

Raise your hand

LIVING LABS

What is a living lab?

A real place where four groups develop and test new solutions together, before they are rolled out.

The city

Sets the questions and opens its streets

Researchers

Develop, measure and evaluate

Companies

Test, adopt and scale solutions

Residents

Take part and give feedback

PURPOSE

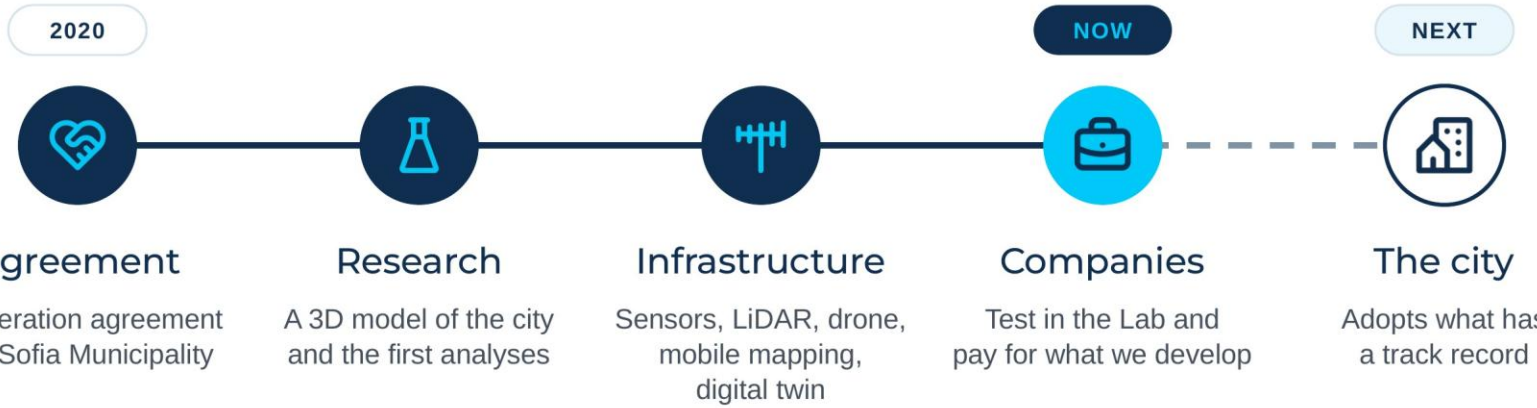
- Test in real life, not on paper
- Learn fast, at low cost
- Scale what works to other cities

GATE INSTITUTE

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How our Lab grew

GATE Institute, Big Data for Smart Society: a research centre for big data and AI in Sofia, founded in 2019.



Research built the expertise and the infrastructure. Companies now build the track record.

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THE INFRASTRUCTURE

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A district that measures itself

Lozenets, Sofia: a real neighbourhood wired as a permanent test bed, together with Sofia Municipality.

12

air-quality stations

Gases, particulates and weather

60

noise sensors

Sound level and its direction

50

pedestrian counters

Crossings, park entrances, public buildings

6

LiDAR sensors

Vehicles and pedestrians at a busy intersection

Drone: photogrammetry and LiDAR scans

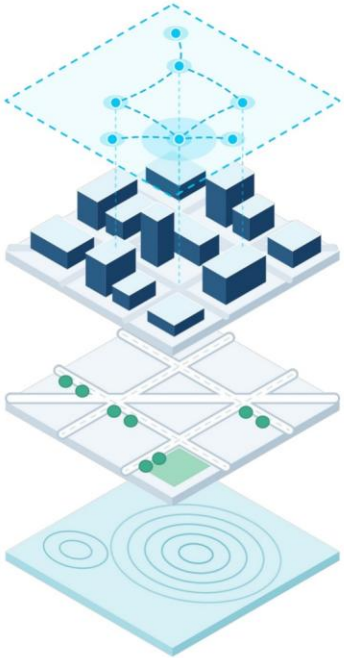
Cooperation agreement with Sofia Municipality

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THE DIGITAL TWIN

One model, many layers

Everything we measure lands in one 3D model of the city, built on open standards.



Live data

Air, noise, pedestrians and traffic, streamed from the Lab's sensors

Buildings

From the cadastre, enriched with point clouds and imagery

Streets & green spaces

Street network, parks and trees

Terrain

Relief, orthophotos and satellite imagery

Open standards **CityGML** **Open APIs**

WHAT'S CHANGING

Companies are coming to the Lab

Companies come to us, and they want to pay for solutions developed in the Lab.

SOLUTIONS WE DEVELOP

- Digital-twin services for sites and districts
- Wind, sun and heat simulation for new developments
- Walkability and accessibility analysis
- LiDAR traffic and pedestrian analytics
- Sensor-network design
- Privacy-first mobility data, in development

A REAL DISTRICT TO TEST IN

Companies bring their own product and test it under real conditions, with independent ground truth from our LiDAR and sensors.

Example: a European startup is validating its camera-based sensing against our intersection LiDAR.

Startups can test and grow on the Lab's infrastructure, and the Lab helps spin-offs take solutions to market.

THREE TAKEAWAYS

Who does what in a living lab

What we learned in Sofia, and would recommend to any city.

1

The city owns it

It owns the questions, ideally the lab itself, and adopts what is proven.

2

Research builds it

Institutes develop innovations for the city that scale to other places and contexts.

3

Business uses and scales it

Companies use and pay for ready solutions. Startups test and grow in the lab; spin-offs take solutions to market.

And residents take part at every step. They are why the lab exists.

WHY THE CITY ADOPTS LAST

Companies use and pay

↓

Solutions earn a track record

↓

The city adopts with less risk

Thank you

CONNEXUS 2026 · Connecting Ideas. Creating Impact.

Let's turn city data into decisions — together.

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Scan to explore the Lab's platforms